Chapter 13, Problem 40P

Problem

The primary of an ideal transformer with a turns ratio of 5 is connected to a voltage source with Thevenin parameters *v*Th = 10 cos 2000*t* V and *R*Th = 100 Ω. Determine the average power delivered to a 200-Ω load connected across the secondary winding.

Step-by-step solution

Step 1 of 6

Consider the primary of an ideal transformer with turns ratio 1is connected to the voltage source $v_{Th} = 10\cos 2000t$ V and $R_{Th} = 200\ \Omega$ as shown in Figure 1.

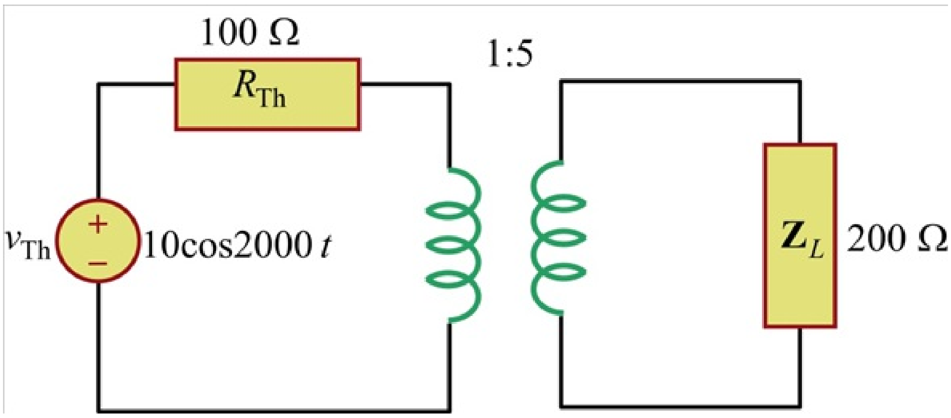

Figure 1

Step 2 of 6

Draw the following circuit diagram the value of Z_{in} input impedance seen at the input side.

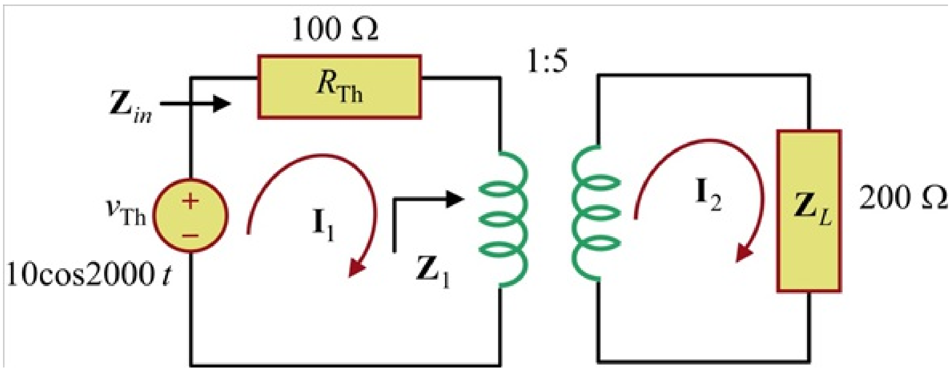

Figure 2

Step 3 of 6

Calculate the value of $\mathbf{Z}_1$.

Write the following expression.

$$\frac{\mathbf{Z}_1}{\mathbf{Z}_L} = \left(\frac{N_1}{N_2}\right)^2$$

$$\mathbf{Z}_1 = \left(\frac{N_1}{N_2}\right)^2 \mathbf{Z}_L$$

Substitute $1:5$ for $\frac{N_1}{N_2}$, $200\ \Omega$ for $\mathbf{Z}_L$ in this expression.

$$\begin{aligned}\mathbf{Z}_1 &= \frac{(200)}{(5)^2} \\ &= 8\end{aligned}$$

Therefore, the value of $\mathbf{Z}_1$ is 8.

Step 4 of 6

Calculate the value of input impedance $\mathbf{Z}_{in}$ seen at the input side.

$$\mathbf{Z}_{in} = \mathbf{Z}_p + \mathbf{Z}_1$$

Substitute $100\ \Omega$ for $\mathbf{Z}_p$, 8 for $\mathbf{Z}_1$ in this expression.

$$\begin{aligned}\mathbf{Z}_{in} &= 100 + 8\ \Omega \\ &= 108\ \Omega\end{aligned}$$

Therefore, the value of $\mathbf{Z}_{in}$ is $108\ \Omega$.

Step 5 of 6

Calculate the value of the current $\mathbf{I}_1$.

Write the expression for the current $\mathbf{I}_1$.

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{\mathbf{Z}_{in}}$$

Substitute 10 V for $\mathbf{V}_1$, $108\ \Omega$ for $\mathbf{Z}_{in}$ in this expression.

$$\begin{aligned}\mathbf{I}_1 &= \frac{10}{108} \\ &= 92.52 \times 10^{-3}\end{aligned}$$

Therefore, the value of $\mathbf{I}_1$ is $92.52\ \text{mA}$.

Step 6 of 6

Calculate the value of the current $\mathbf{I}_2$.

Write the expression for the current $\mathbf{I}_2$.

$$\frac{\mathbf{I}_2}{\mathbf{I}_1} = \frac{N_1}{N_2}$$
$$\mathbf{I}_2 = \left(\frac{N_1}{N_2}\right)\mathbf{I}_1$$

Substitute $1:5$ for $\frac{N_1}{N_2}$, 92.52 mA for $\mathbf{I}_1$ in this expression.

$$\begin{aligned}\mathbf{I}_2 &= \left(\frac{1}{5}\right)92.52 \text{ mA}\\ &= 18.504 \text{ mA}\end{aligned}$$

Therefore, the value of $\mathbf{I}_2$ is 18.504 mA.

Calculate the value of P_{avg} average power transferred to the load $\mathbf{Z}_L$.

Write the expression for the value of P_{avg} average power transferred to the load $\mathbf{Z}_L$.

$$P_{\text{avg}} = \frac{1}{2}\left|\mathbf{I}_2\right|^2 \mathbf{Z}_L$$
$$\begin{aligned}P_{\text{avg}} &= \frac{1}{2}\left|18.504 \text{ mA}\right|^2 (200)\\ &= 100\left(342.39\times10^{-6}\right)\\ &= 34.3\times10^{-3}\end{aligned}$$

Therefore, the value of P_{avg} is $\boxed{34.3 \text{ mW}}$.